Alibaba E-commerce database management

Queries Implementation

1. Basic SQL Queries

1.1 SELECT Query

Display all products

```
PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       101 JBL Earbuds                                                 1
      2299         10 Wholesale
Bluetooth wireless earbuds
       250         60


PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       102 Bluetooth Speaker                                           1
      3999          5 Wholesale
Portable wireless speaker
       180         30
```

```
PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       103 Wireless Headphones                                         1
      5999          5 Wholesale
Noise cancelling headphones
       120         20


PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       104 Cotton Shirt                                                2
       999         10 Wholesale
Comfortable cotton shirt
       150         40
```

```
PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       105 Denim Jeans                                                 2
      1499         10 Wholesale
Blue denim jeans
       100         25


PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       106 Python Programming Book                                     3
       699          5 Wholesale
Python programming guide
        80         15
```

```
PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       107 DBMS Fundamentals                                           3
       799          5 Wholesale
Database management book
        70         20


PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       108 Mixer Grinder                                               4
      3499          5 Wholesale
Kitchen mixer grinder
        60         10
```

```
PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       109 Electric Kettle                                             4
      1999          5 Wholesale
Stainless steel electric kettle
        90         18


PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       110 Football                                                    5
       899         10 Wholesale
Professional football
       130         35


10 rows selected.
```

Display selected product details

```
PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
       101 JBL Earbuds                                          2299
       102 Bluetooth Speaker                                    3999
       103 Wireless Headphones                                  5999
       104 Cotton Shirt                                          999
       105 Denim Jeans                                          1499
       106 Python Programming Book                               699
       107 DBMS Fundamentals                                     799
       108 Mixer Grinder                                        3499
       109 Electric Kettle                                      1999
       110 Football                                              899

10 rows selected.
```

1.2 WHERE clause

Display table product where price is less than 1499

```
SQL> select*from product where price<1499;

PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       104 Cotton Shirt                                               2
       999         10 Wholesale
Comfortable cotton shirt
       150         40


PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       106 Python Programming Book                                    3
       699          5 Wholesale
Python programming guide
        80         15
```

```
PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       107 DBMS Fundamentals                                             3
       799          5 Wholesale
Database management book
        70         20


PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       110 Football                                                      5
       899         10 Wholesale
Professional football
       130         35
```

Display product table where price is higher than 4000

```
SQL> select*from product where price > 4000;

PRODUCT_ID PRODUCT_NAME                                        CATEGORY_ID
---------- -------------------------------------------------- -----------
     PRICE        MOQ LISTING_TYPE
---------- ---------- --------------------
KEY_FEATURES
--------------------------------------------------------------------------------
UNITS_SOLD      STOCK
---------- ----------
       103 Wireless Headphones                                           1
      5999          5 Wholesale
Noise cancelling headphones
       120         20
```

1.3 ORDER BY

Display price in ascending order

```
SQL> select product_id,product_name,price from product order by price asc;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
       106 Python Programming Book                                   699
       107 DBMS Fundamentals                                         799
       110 Football                                                  899
       104 Cotton Shirt                                              999
       105 Denim Jeans                                              1499
       109 Electric Kettle                                          1999
       101 JBL Earbuds                                              2299
       108 Mixer Grinder                                            3499
       102 Bluetooth Speaker                                        3999
       103 Wireless Headphones                                      5999

10 rows selected.
```

Display price in descending order

```
SQL> select product_id,product_name,price from product order by price desc;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
       103 Wireless Headphones                                      5999
       102 Bluetooth Speaker                                        3999
       108 Mixer Grinder                                            3499
       101 JBL Earbuds                                              2299
       109 Electric Kettle                                          1999
       105 Denim Jeans                                              1499
       104 Cotton Shirt                                              999
       110 Football                                                  899
       107 DBMS Fundamentals                                         799
       106 Python Programming Book                                   699

10 rows selected.
```

1.4 DISTINCT

Distinct the price and display it.

```
SQL> select distinct price from product;

     PRICE
----------
      2299
      3999
      5999
       999
      1499
       699
       799
      3499
      1999
       899

10 rows selected.
```

2.Search products

2.1 Search products based on price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 1499 and 2999;

PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
       101 JBL Earbuds                                          2299
       105 Denim Jeans                                          1499
       109 Electric Kettle                                      1999
```

2.2 Search products based on category

```
SQL> SELECT Product_ID, Product_Name, Category_ID
  2  FROM Product
  3  WHERE Category_ID = 3;

PRODUCT_ID PRODUCT_NAME                                    CATEGORY_ID
---------- -------------------------------------------------- -----------
       106 Python Programming Book                                  3
       107 DBMS Fundamentals                                        3
```

3. Retrieve information

3.1 Display all in Customer

```
SQL> SELECT Customer_ID, Customer_Name, Email, Contact_No
  2  FROM Customer;

CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              CONTACT_NO
-------------------------------------------------- ----------
        101 ABC Traders
abc@gmail.com                                      9876543210

        102 Tech Solutions
tech@gmail.com                                     9876543211

        103 Green Mart
green@gmail.com                                    9876543212


CUSTOMER_ID CUSTOMER_NAME
----------- --------------------------------------------------
EMAIL                                              CONTACT_NO
-------------------------------------------------- ----------
        104 Eco Products
eco@gmail.com                                      9876543213

        105 Smart Store
smart@gmail.com                                    9876543214
```

3.2 Search for a particular customer

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_ID = 101;

CUSTOMER_ID CUSTOMER_NAME                                          CONTACT_NO
----------- -------------------------------------------------- ----------
EMAIL
--------------------------------------------------
COUNTRY
------------------------------
ADDRESS
--------------------------------------------------
        101 ABC Traders                                        9876543210
abc@gmail.com
India
Chennai
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price < 2000
  4  AND Category_ID = 3;

PRODUCT_ID PRODUCT_NAME                                            PRICE
---------- -------------------------------------------------- ----------
       106 Python Programming Book                                   699
       107 DBMS Fundamentals                                         799
```

4.2 OR Condition

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Category_ID = 1
  4  OR Category_ID = 5;

PRODUCT_ID PRODUCT_NAME                                          PRICE
---------- -------------------------------------------------- ----------
       101 JBL Earbuds                                          2299
       102 Bluetooth Speaker                                    3999
       103 Wireless Headphones                                  5999
       110 Football                                              899
```

4.3 IN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Category_ID IN (1, 2, 3);

PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
       101 JBL Earbuds                                          2299
       102 Bluetooth Speaker                                    3999
       103 Wireless Headphones                                  5999
       104 Cotton Shirt                                          999
       105 Denim Jeans                                          1499
       106 Python Programming Book                               699
       107 DBMS Fundamentals                                     799

7 rows selected.
```

4.4 BETWEEN Operator

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 2999 and 4999;

PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
       102 Bluetooth Speaker                                    3999
       108 Mixer Grinder                                        3499
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID PRODUCT_NAME                                         PRICE
---------- -------------------------------------------------- ----------
       103 Wireless Headphones                                  5999
       102 Bluetooth Speaker                                    3999
       108 Mixer Grinder                                        3499
       101 JBL Earbuds                                          2299
       109 Electric Kettle                                      1999
       105 Denim Jeans                                          1499
       104 Cotton Shirt                                          999
       110 Football                                              899
       107 DBMS Fundamentals                                     799
       106 Python Programming Book                               699

10 rows selected.
```

5.2 Customer Order Report

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      o.Order_ID,
  5      o.Order_Date,
  6      o.Total_Amount
  7  FROM Customer c
  8  JOIN Orders o
  9  ON c.Customer_ID = o.Customer_ID
 10  ORDER BY o.Order_Date DESC;

CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        105 Smart Store                                          1010
10-SEP-26        28900

        104 Eco Products                                         1009
09-SEP-26        15400

        103 Green Mart                                           1008
08-SEP-26        36500
```

```
CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        102 Tech Solutions                                       1007
07-SEP-26        21500

        101 ABC Traders                                          1006
06-SEP-26         9800

        105 Smart Store                                          1005
05-SEP-26        45000


CUSTOMER_ID CUSTOMER_NAME                                      ORDER_ID
----------- -------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        104 Eco Products                                         1004
04-SEP-26        12750

        103 Green Mart                                           1003
03-SEP-26        32000

        102 Tech Solutions                                       1002
02-SEP-26        18500
```

```
CUSTOMER_ID CUSTOMER_NAME                                          ORDER_ID
----------- ---------------------------------------------------- ----------
ORDER_DAT TOTAL_AMOUNT
--------- ------------
        101 ABC Traders                                              1001
01-SEP-26        25000


10 rows selected.
```

5.3 Product Order Report

```
SQL> SELECT
  2      od.Order_ID,
  3      od.Product_ID,
  4      od.Quantity,
  5      od.Unit_Price,
  6      (od.Quantity * od.Unit_Price) AS Subtotal
  7  FROM Order_Details od
  8  ORDER BY Subtotal DESC;

  ORDER_ID PRODUCT_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ---------- ----------
      1005        305         10       4500      45000
      1005        305         10       4500      45000
      1001        301          8       5000      40000
      1008        308         10       3650      36500
      1008        308         10       3650      36500
      1003        303          8       4000      32000
      1003        303          8       4000      32000
      1007        307          7       4300      30100
      1010        310         17       1700      28900
      1010        310         17       1700      28900
      1001        301          5       5000      25000
```

```
  ORDER_ID PRODUCT_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ---------- ----------
      1007        307          5       4300      21500
      1002        302         10       1850      18500
      1002        302         10       1850      18500
      1009        309          7       2200      15400
      1009        309          7       2200      15400
      1004        304         15        850      12750
      1006        306          7       1400       9800
      1006        306          7       1400       9800
      1004        304          8        900       7200

20 rows selected.
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      COUNT(o.Order_ID) AS Total_Orders
  5  FROM Customer c
  6  JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.Customer_Name
  9  ORDER BY Total_Orders DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_ORDERS
----------- -------------------------------------------------- ------------
        101 ABC Traders                                                 2
        102 Tech Solutions                                              2
        105 Smart Store                                                 2
        104 Eco Products                                                2
        103 Green Mart                                                  2
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      c.Customer_ID,
  3      c.Customer_Name,
  4      SUM(o.Total_Amount) AS Total_Purchase
  5  FROM Customer c
  6  JOIN Orders o
  7  ON c.Customer_ID = o.Customer_ID
  8  GROUP BY c.Customer_ID, c.Customer_Name
  9  ORDER BY Total_Purchase DESC;

CUSTOMER_ID CUSTOMER_NAME                                      TOTAL_PURCHASE
----------- -------------------------------------------------- --------------
        105 Smart Store                                             73900
        103 Green Mart                                              68500
        102 Tech Solutions                                          40000
        101 ABC Traders                                             34800
        104 Eco Products                                            28150
```